

STAT 341: Tutorial Test 2 - Sample 1

First name

Last name

Signature

ID Number

Email

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

Please read: this practice test is designed to give you a clear sense of the sort of things that can come up in exams. Please note that this is, of course, only a subset of the material that could be covered. I would **strongly** encourage you to attempt this test under exam conditions - do not be tempted to look up something in your notes until afterwards! This is one reason why it is common for students to find real exams much harder than practice ones, even though the material is of an equivalent difficulty!

1. Consider estimating the mean of the population $\mathcal{P} = \{1, 2, 3, 4\}$ based on a random sample of size 2 (without replacement). The sampling protocol (probability of all possible samples) is summarized in the table below.

$\mathcal{S}$	$\{1, 2\}$	$\{1, 3\}$	$\{1, 4\}$	$\{2, 3\}$	$\{2, 4\}$	$\{3, 4\}$
$P(\mathcal{S})$	0.15	0.25	0.1	0	0.2	0.3

- (a) [**2 marks**] What is sampling bias for the sample mean, $\bar{Y}$.

- (b) [**2 marks**] Calculate the Horvitz-Thompson estimate of the population mean based on the sample $\mathcal{S} = \{2, 4\}$.

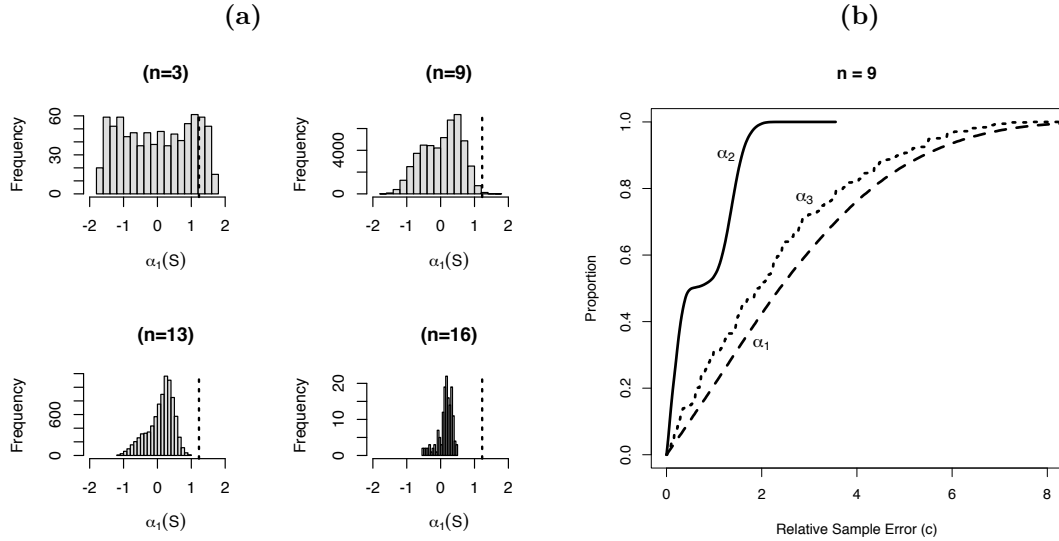
2. [2 marks] Suppose we are interested in the average length of sharks in great white shark encounters with humans in US waters but have access only to those encounters which were in Australian waters.

- The **target population** is

- The **study population** is

3. [2 marks] Every measuring system has sources of potential error. Name two sources of potential measurement error.

4. Consider a population $\mathcal{P} = \{u_1, \dots, u_{18}\}$ from which a simple random sample of size n without replacement is taken. Interest lies in studying the skewness in this population. Based on all possible samples of sizes $n = 3$, $n = 9$, $n = 13$, and $n = 16$, panel (a) plots histograms of a skewness attribute $\alpha_1(\mathcal{S})$ for different sample sizes. Superimposed in black dashed lines on the histograms is the population value $\alpha_1(\mathcal{P})$. Beside α_1 , two other skewness attributes α_2 and α_3 exist in statistics literature. Panel (b) plots the consistency curves for these three attributes. In particular, given a value c , each curve represents the proportion of samples with relative sample error below c .



(a) [2 marks] What do the plots in panel (a) tell you about the estimator α_1 ?

(b) [2 marks] What do the plots in panel (b) tell you about the estimators α_1 , α_2 and α_3 ?

5. (a) [4 marks] Describe in point form a test of significance using a discrepancy measure $D(\mathcal{P}_1, \mathcal{P}_2)$

(b) [1 mark] Why is $D(\mathcal{P}_1, \mathcal{P}_2) = |a(\mathcal{P}_1) - a(\mathcal{P}_2)|$ a discrepancy measure?

6. **[6 marks (1 mark each)]** In the statements below, complete the sentence or determine if the statement is true or false.

- (a) There exist a total of possible ordered samples, each of size 3, when taking a simple random sample with replacement from the population $\mathcal{P} = \{u_1, u_2, u_3, u_4, u_5, u_6\}$.
- (b) The Horvitz-Thompson estimator of the population mean is a scale equivariant estimator.
 - i. True
 - ii. False
- (c) The difference between the attribute of the study population and that of the sample is called the study error.
 - i. True
 - ii. False
- (d) If the sampling protocol is simple random sampling without replacement, then the sample average is also the Horvitz-Thompson estimator of population mean.
 - i. True
 - ii. False
- (e) The relative error of an estimator is a number between 0 and 1.
 - i. True
 - ii. False
- (f) Scale invariance is a desirable property for measures of spread in a population.
 - i. True
 - ii. False